Digital epidemiology

Lesson 6

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FONDAZIONE BRUNO KESSLER

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Mathematical modeling of epidemics

Modeling

reality

abstraction,
conceptualization

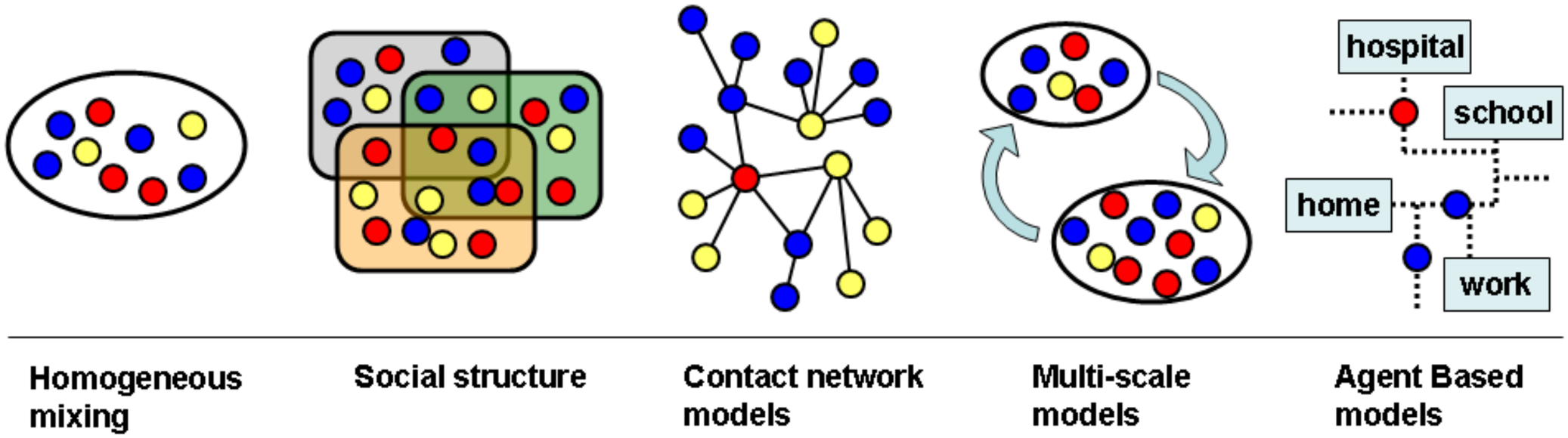
“

All models are wrong.

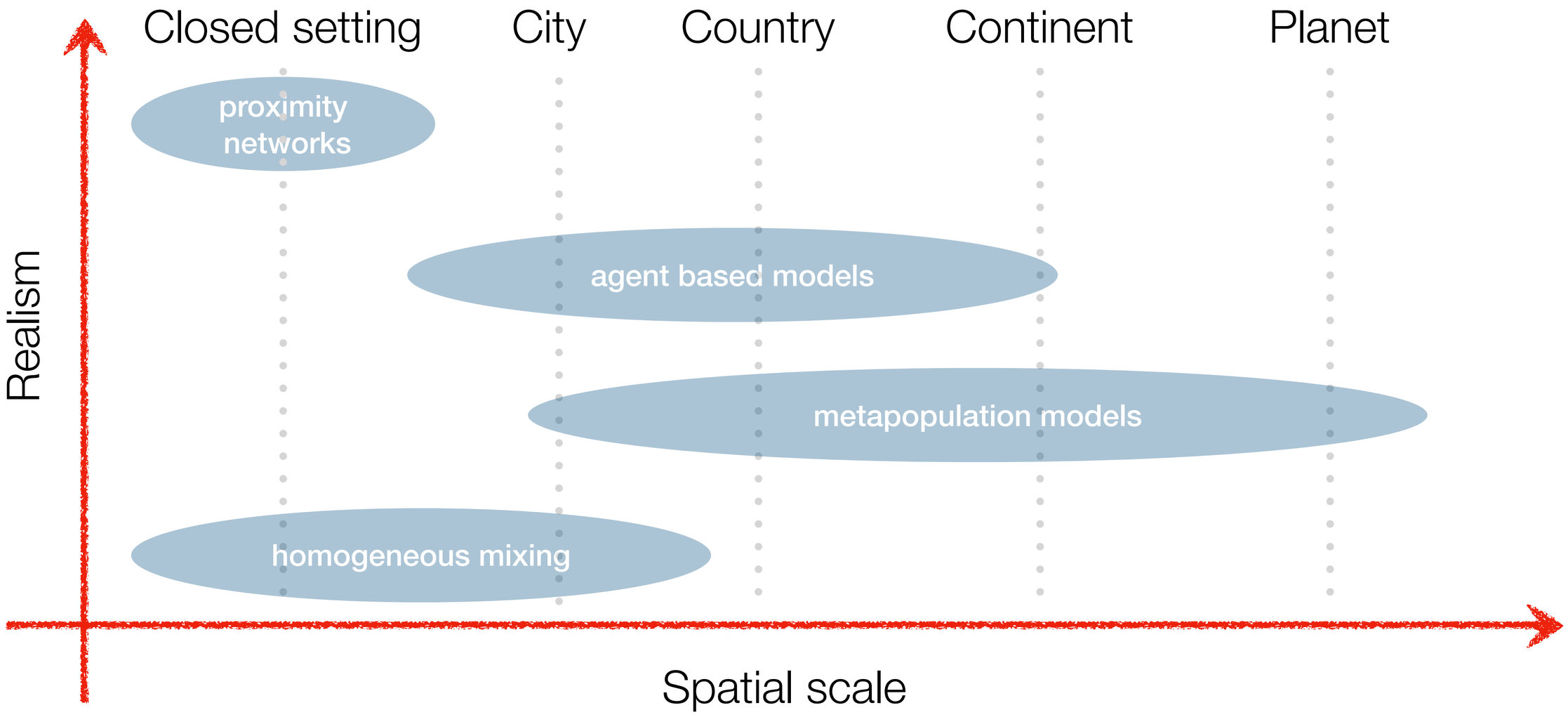
Some are useful.

-George E. P. Box

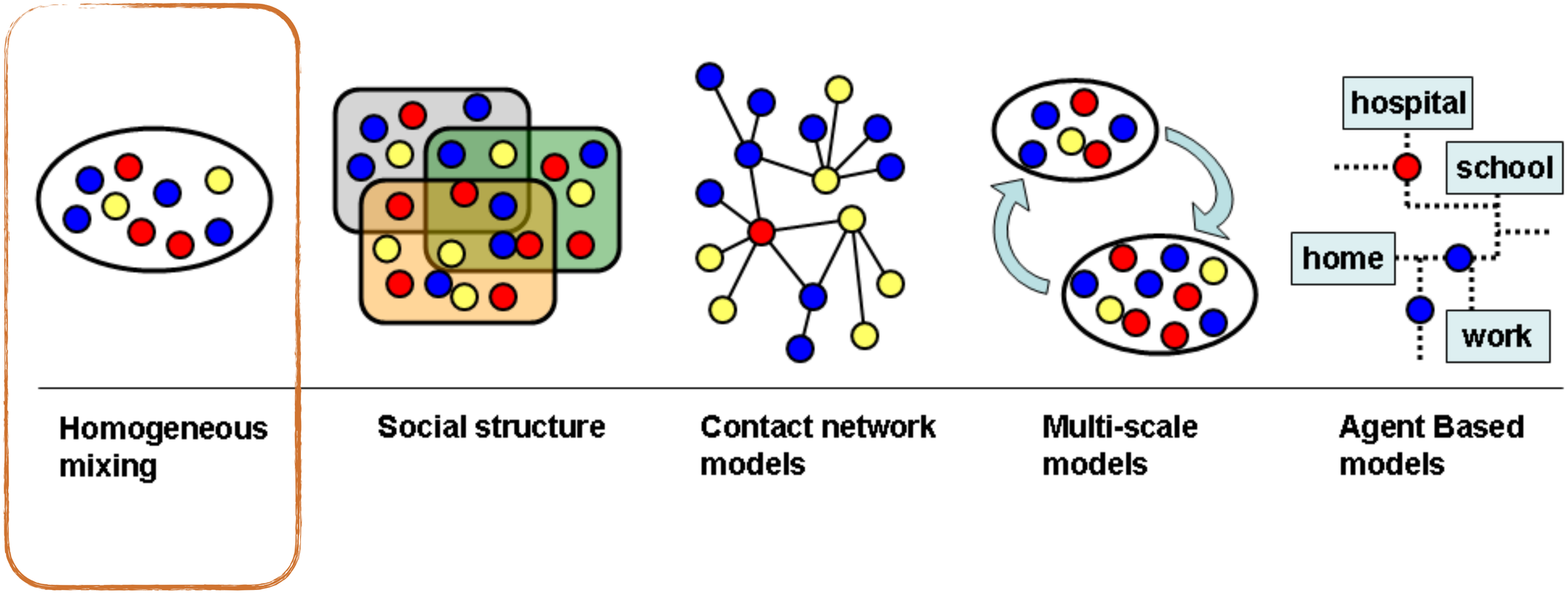
Models


hospital
school
home
work
Homogeneous mixing
Social structure
Contact network models
Multi-scale models
Agent Based models

Spatial scales



Models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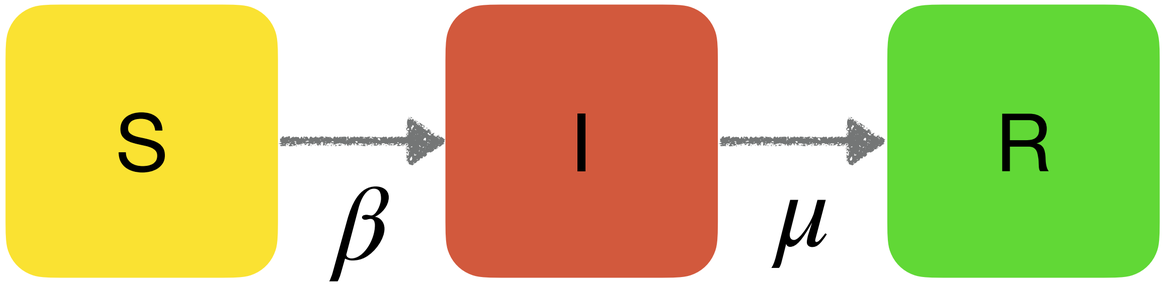
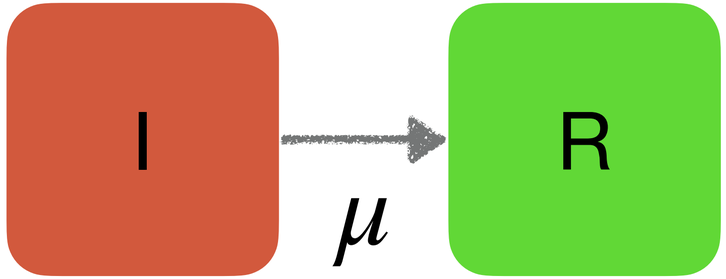
The SIR model



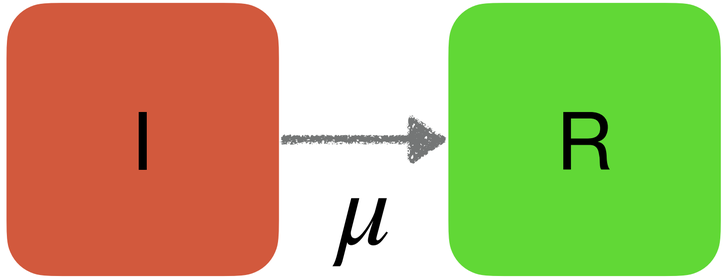
- Population is closed and constant (no demographics) $S + I + R = N$
- Population is “well mixed” (no heterogeneities)
- **Recovery** takes place at constant rate: μ
- **Transmission** rate is constant: β

Recovery transition



- The transition from I to R is **spontaneous**. It does not depend on other compartments.
- The model assumes a constant probability of recovery in the time unit, μ
- In a discrete-time formulation, the time an individual will spend on average in the infectious compartment, the **mean infectious period**, is then equal to μ^{-1} time steps.

Recovery transition



- In a continuous-time formulation, μ is a rate

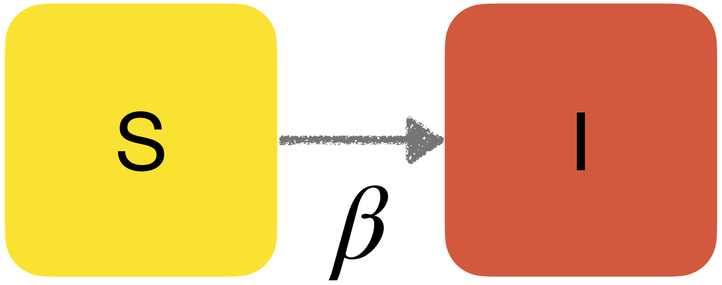
$$P_{inf}(\tau) = \mu e^{-\mu\tau}$$

Probability an individual spends τ in the infectious compartment

$$\langle \tau \rangle = \mu^{-1}$$

Average infectious period

Transmission



- **Homogeneous mixing**: individuals interact at random with each other.
- We define the *force of infection* as the probability that an individual S may contract the infection in a single time step

$$\lambda = \beta \frac{I}{N}$$

Transmission

$$\lambda = \beta \frac{I}{N}$$

frequency dependent

$$\beta = \hat{\beta}\kappa$$

κ → number of contacts

$\hat{\beta}$ → transmission per contact

$$\boxed{\Delta S = \lambda S = \beta \frac{SI}{N} \Delta t}$$

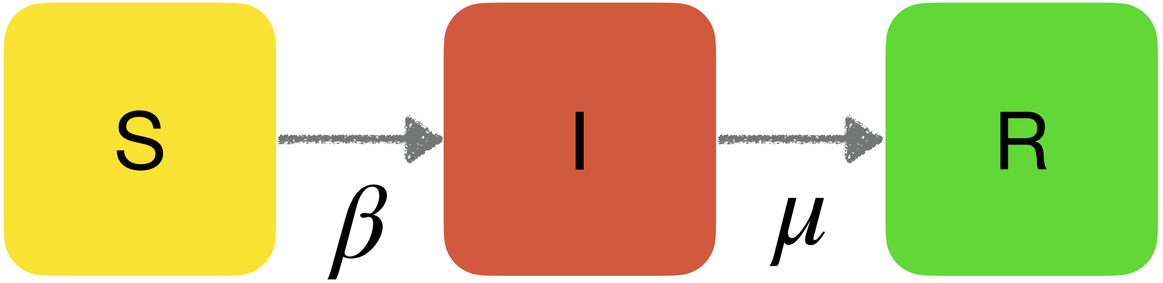
The SIR model

$$S(t+\Delta t) - S(t) = -\beta \frac{SI}{N} \Delta t$$

$$I(t+\Delta t) - I(t) = \beta \frac{SI}{N} \Delta t - \mu I \Delta t$$

frequency dependent

$$R(t+\Delta t) - R(t) = \mu I \Delta t$$



The SIR model

$$\begin{cases} \dfrac{dS}{dt} = -\beta \dfrac{SI}{N} \\ \dfrac{dI}{dt} = \beta \dfrac{SI}{N} - \mu I \\ \dfrac{dR}{dt} = \mu I \end{cases}$$

- **No closed solution**
- We can solve the system in the **early stage approximation**
- Two fundamental results from the model:
  1. The **threshold phenomenon**
  2. The **asymptotic state**, i.e. why the epidemic dies out

Epidemic threshold

$$\begin{cases} \dfrac{dS}{dt} = -\beta \dfrac{SI}{N} \\ \dfrac{dI}{dt} = \beta \dfrac{SI}{N} - \mu I \\ \dfrac{dR}{dt} = \mu I \end{cases}$$

Early stage approximation

$$t \to 0$$

$$I(0) \ll N$$

$$S(0) \simeq N$$

Epidemic threshold

Early stage approximation

$$t \to 0$$

$$I(0) \ll N$$

$$S(0) \simeq N$$

$$\frac{dI}{dt} \simeq (\beta - \mu)I$$

$$\frac{dI}{I} \simeq (\beta - \mu)dt$$

Epidemic threshold

Early stage approximation

$$t \to 0$$

$$I(0) \ll N$$

$$S(0) \simeq N$$

$$I(t) \simeq I_0 e^{(\beta-\mu)t}$$

$$\beta - \mu > 0$$

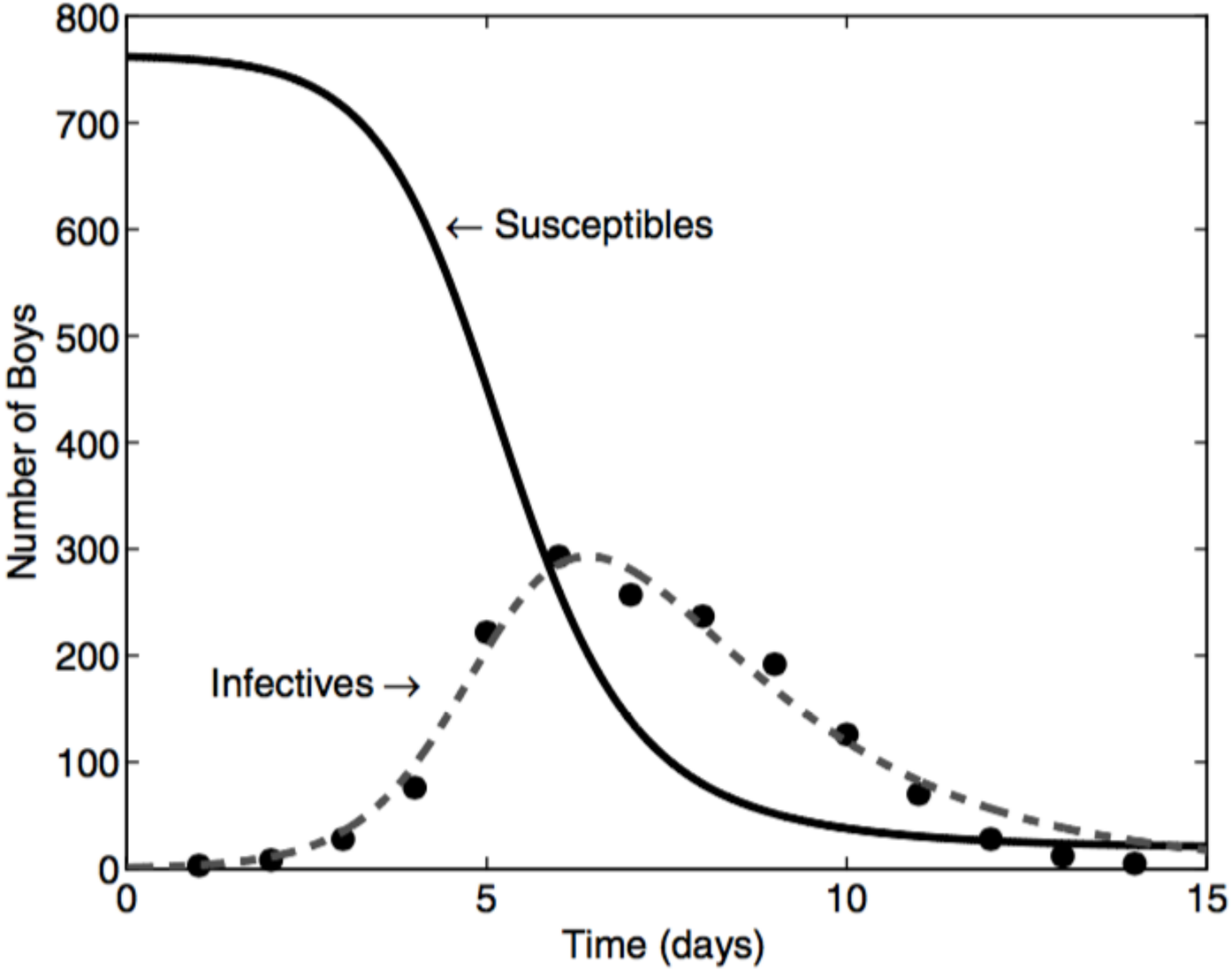
$$R_0 = \frac{\beta}{\mu} > 1$$

Epidemic threshold

- The **basic reproductive number** is the average number of individuals that are infected by an infectious one during his infectious period, in a fully susceptible population.
- An epidemic outbreak **can occur only if** the basic reproductive number is **larger than 1**.

SIR application

- Flu epidemic in a boarding school in England, 1978 (data from BMJ)
- Data can be fitted with SIR by ordinary least squares
- Estimated parameters:
  - $R_0 = 3.65$
  - infectious period = 2.2 days



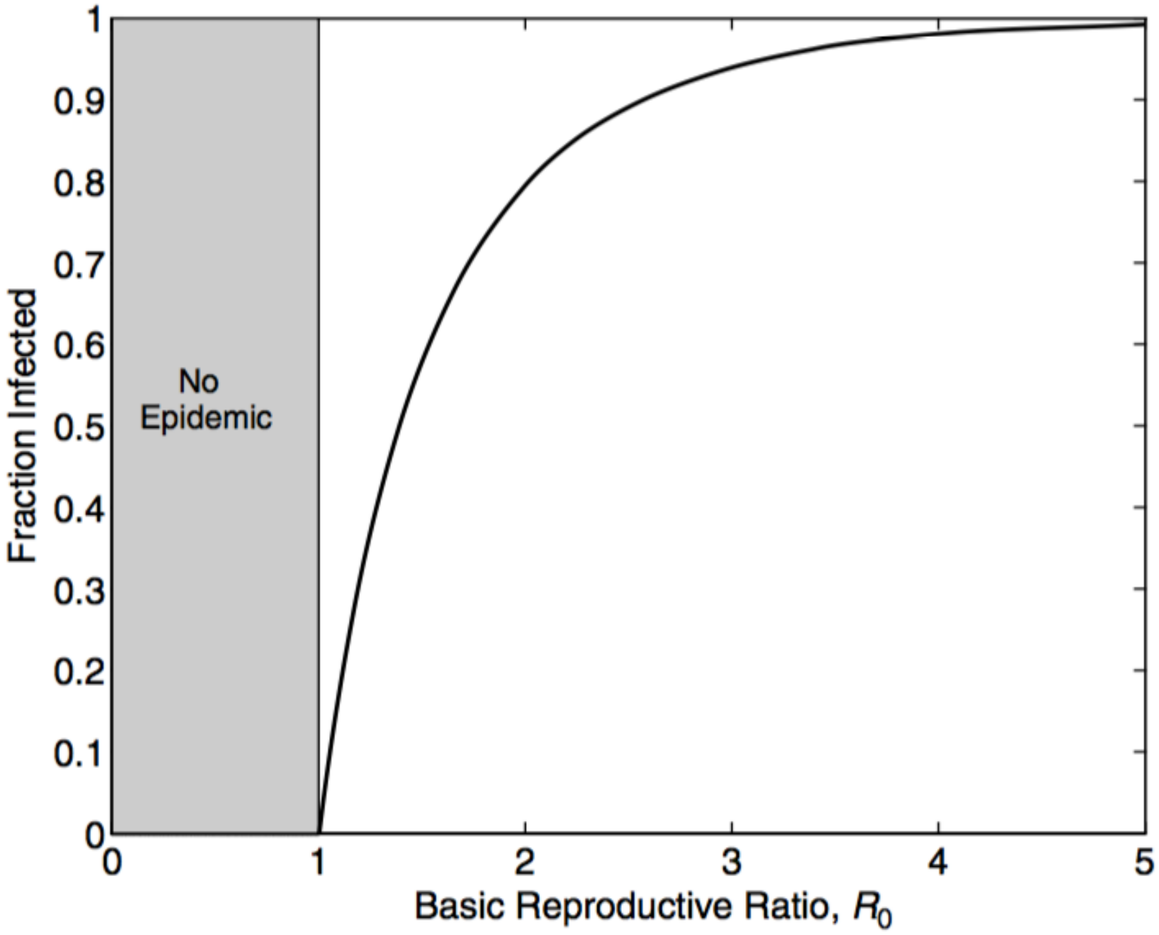
The asymptotic state

$$\begin{cases} \dfrac{dS}{dt} = -\beta \dfrac{SI}{N} \\ \dfrac{dI}{dt} = \beta \dfrac{SI}{N} - \mu I \\ \dfrac{dR}{dt} = \mu I \end{cases}$$

$$\frac{dS}{dR} = -R_0 \frac{S}{N}$$

$$\boxed{s(t) = s(0)e^{-R_0 r(t)}}$$

The asymptotic state



$$r(\infty) = 1 - s(\infty) = 1 - s(0)e^{-R_0 r(\infty)}$$

$$1 - r(\infty) - s(0)e^{-R_0 r(\infty)} = 0$$

There is a non-zero solution for r if $R_0>1$.

The chain of transmission eventually breaks due to the decline in infectives, **not due to a complete lack of susceptibles**.

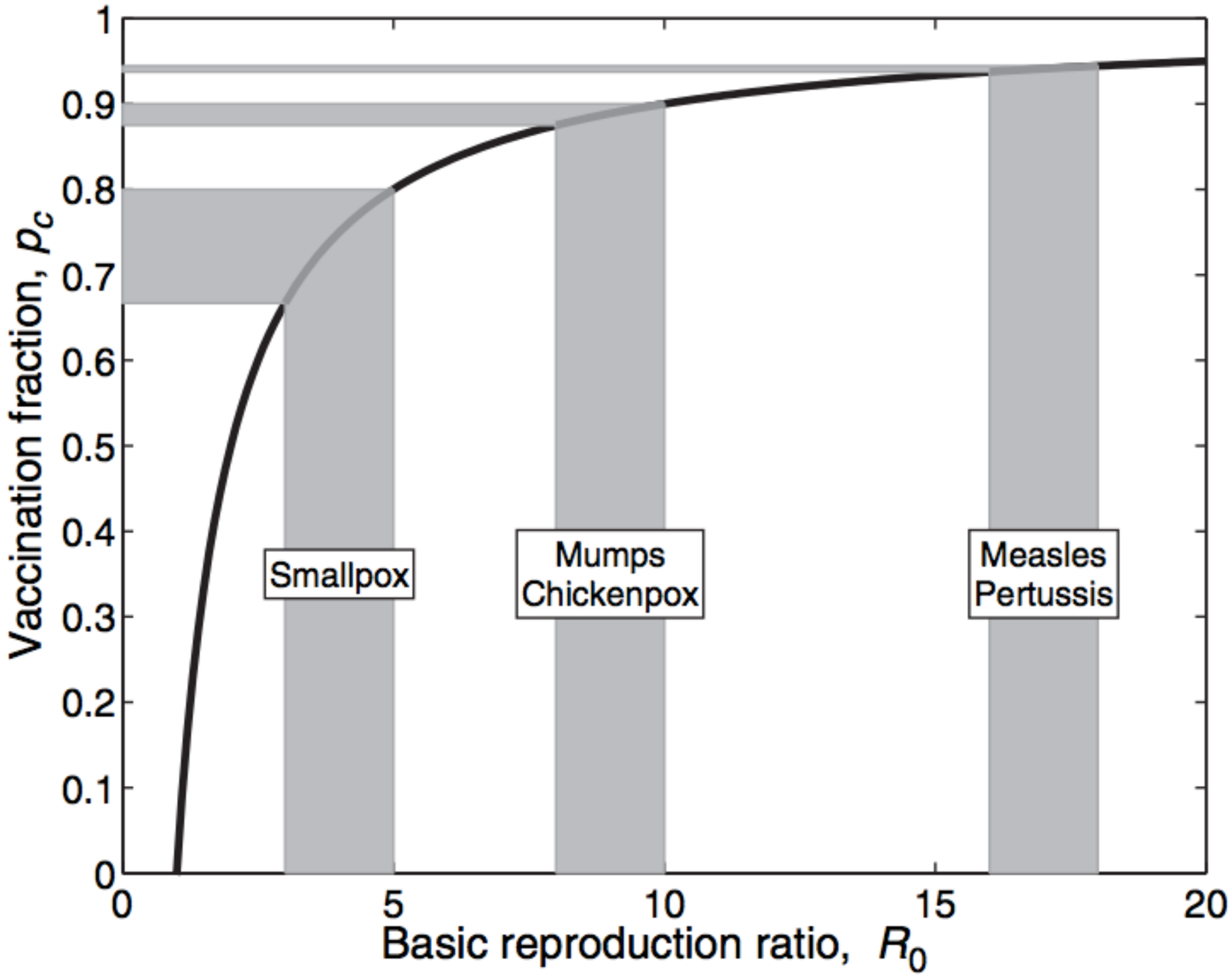
The herd immunity

- We can stop the epidemic by reducing the pool of susceptible through removal or **vaccination**.

$$\frac{dI}{dt} \simeq \beta(1 - f_V)I - \mu I$$

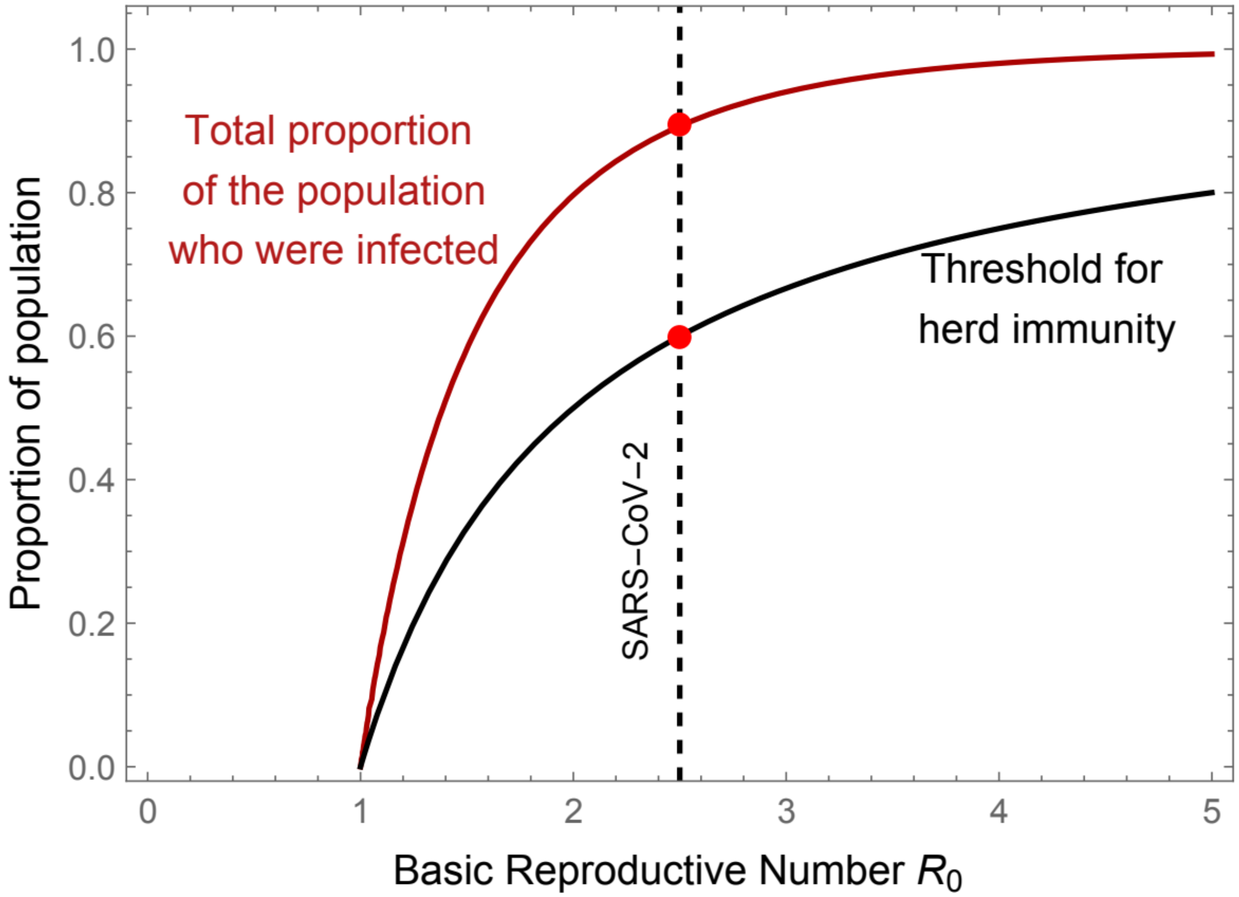
$$\frac{\beta}{\mu}(1 - f_V) > 1 \quad \longrightarrow \quad f_{V,th} = 1 - \frac{1}{R_0}$$

The herd immunity

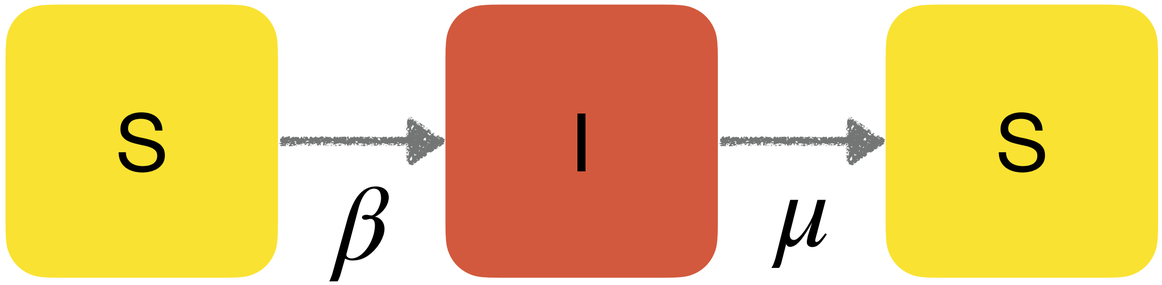


The overshoot

- What if we just let the epidemic spread through the population until we reach herd immunity? There will be many more infections than the threshold.



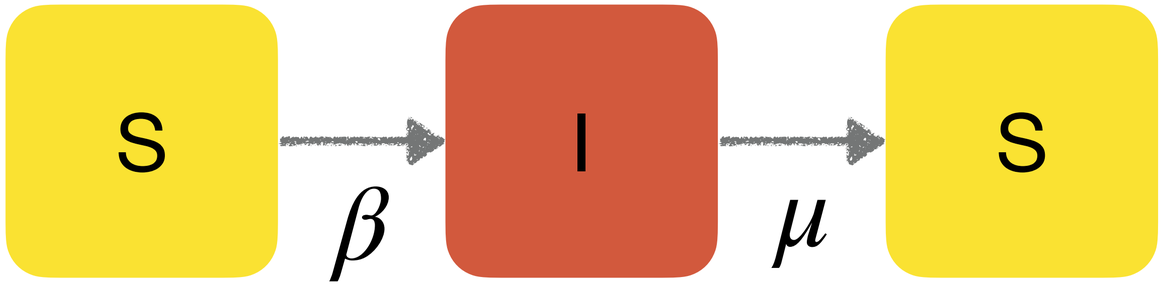
The SIS model



$$\begin{cases} \dfrac{dS}{dt} = -\beta \dfrac{SI}{N} + \mu I \\ \dfrac{dI}{dt} = \beta \dfrac{SI}{N} - \mu I \end{cases}$$

$$S + I = N$$

The SIS model



$$\frac{dI}{dt} = (\beta - \mu)I - \frac{\beta}{N}I^2$$

Logistic equation

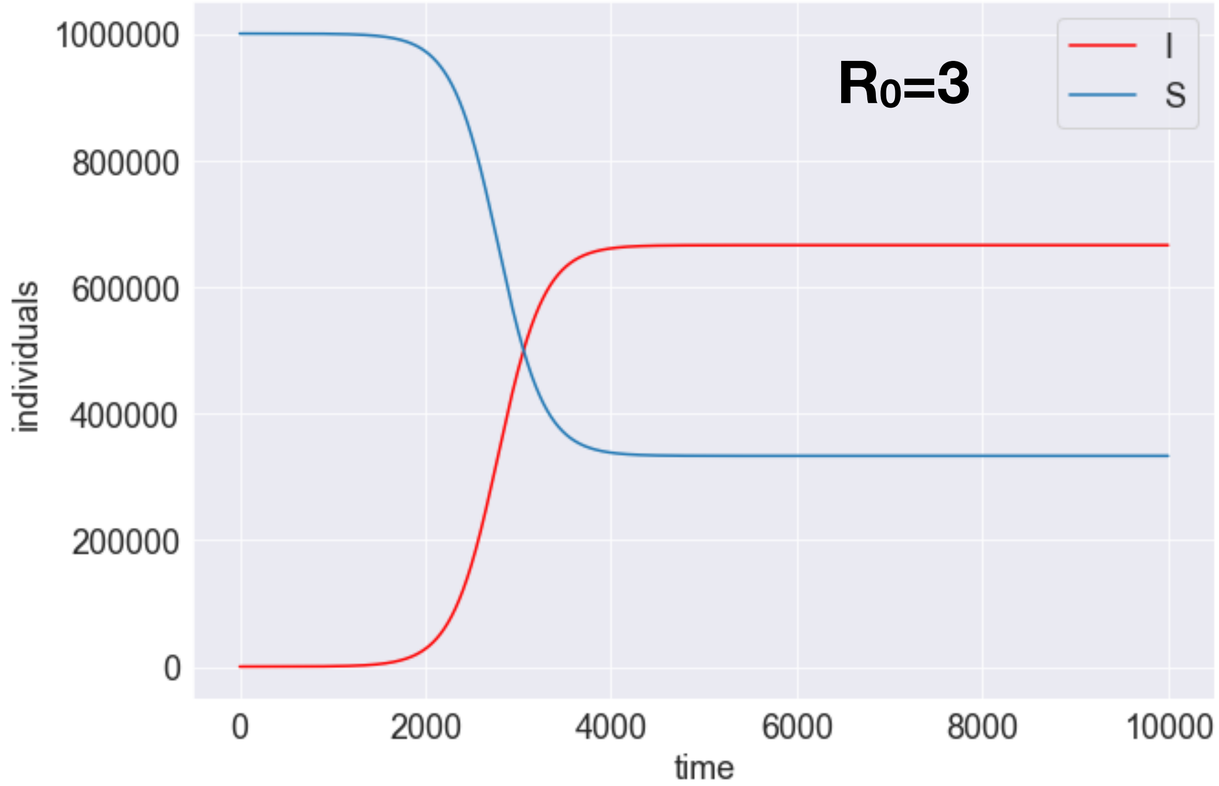
$$\frac{\beta}{\mu} > 1 \Rightarrow \lim_{t \to +\infty} I(t) = \left(1 - \frac{\mu}{\beta}\right) N$$

Non-trivial equilibrium

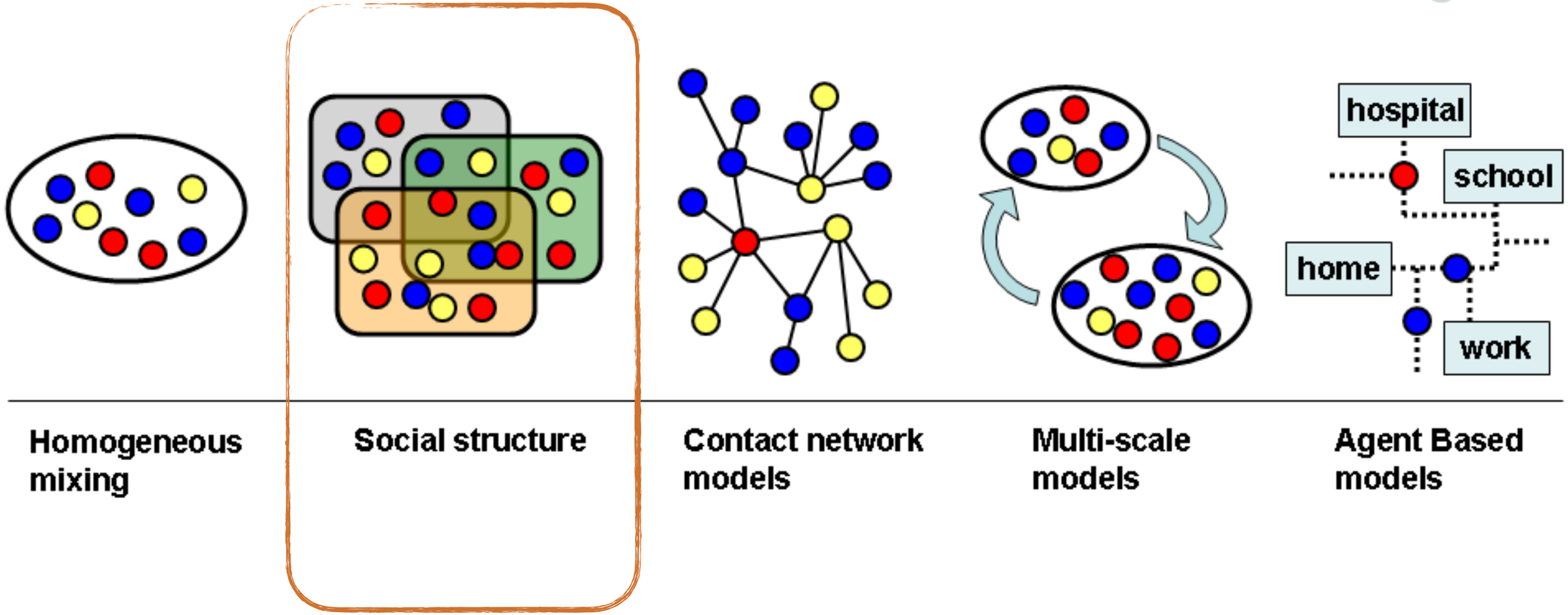
The SIS model

- **The disease persists as long as $R_0>1$**
- **The system reaches an endemic state, with:**

$$I^* = \left(1 - \frac{1}{R_0}\right) N$$



Models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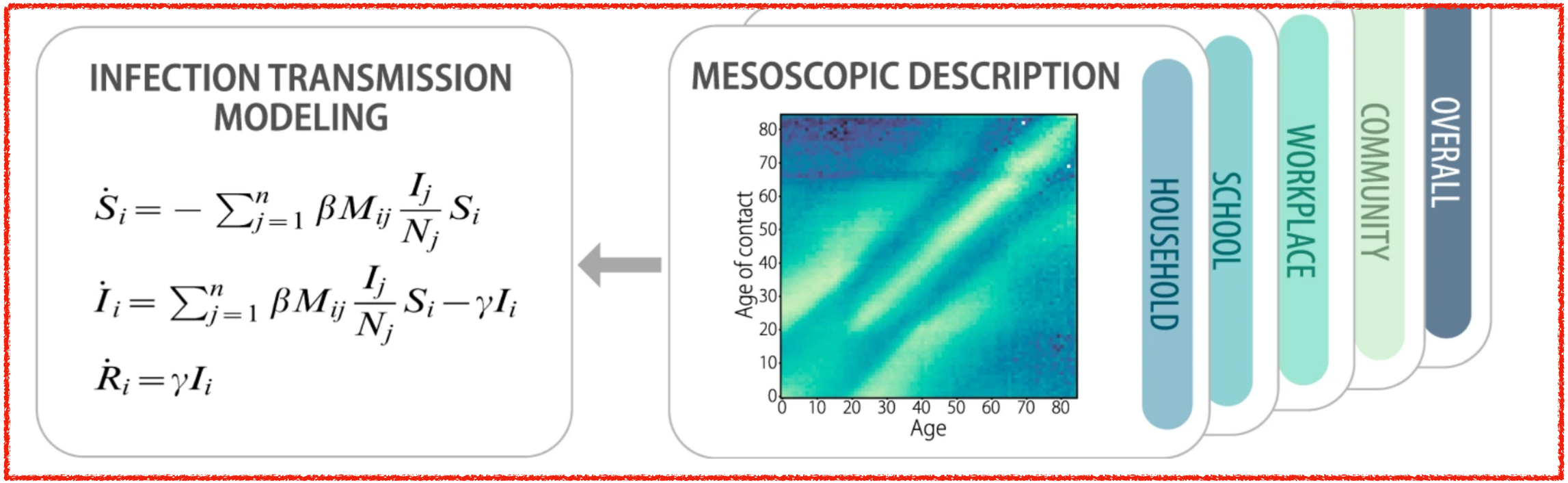
Age-structured models



- Compartments are structured into n age classes
- M_{ij} represents the average contact rate between individuals of age i and j

Contact matrices

- Contact matrices can be estimated in different ways
- Through **empirical surveys,** which are more accurate but require significant resources (Mossong et al. 2008).
- By the creation of **synthetic populations** (Fumanelli et al. 2012).

PLOS COMPUTATIONAL BIOLOGY

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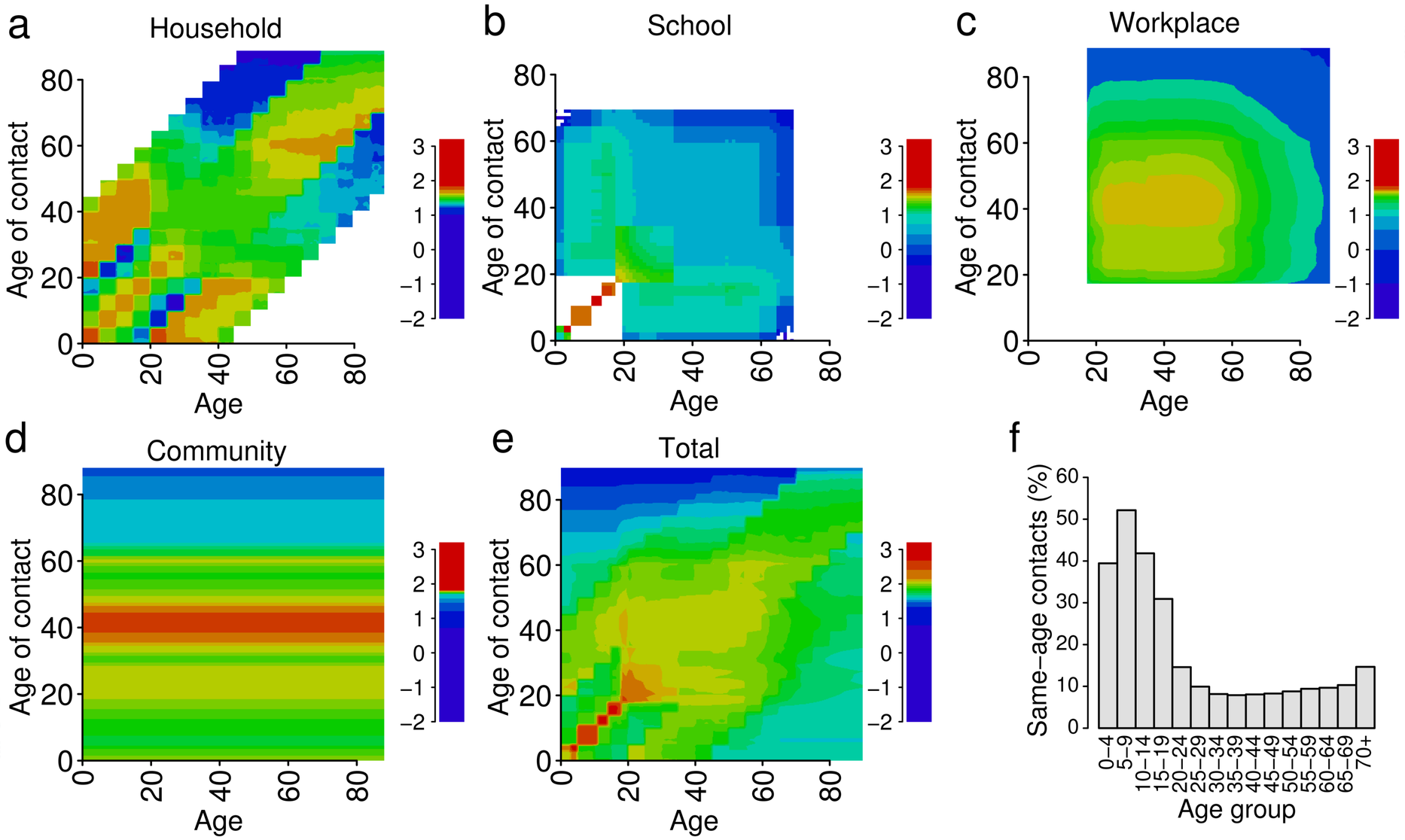
RESEARCH ARTICLE

Inferring the Structure of Social Contacts from Demographic Data in the Analysis of Infectious Diseases Spread

Laura Fumanelli, Marco Ajelli, Piero Manfredi, Alessandro Vespignani, Stefano Merler

Published: September 13, 2012 • https://doi.org/10.1371/journal.pcbi.1002673

Synthetic populations



Next generation matrix

- In the case of an age-structured epidemic model, what is the value of R_0?
- We can compute it using the so called **“next-generation matrix” approach**, first introduced by Diekmann *et al.* (1990).
- We will not go through a formal proof of the approach but in the case of a simple age-structured SIR model, it is quite intuitive.
- The **basic reproductive number** of the model corresponds to the largest of the absolute values of the eigenvalues of the contact matrix M (**the spectral radius**).

$$R_0 = \rho\left(\frac{\beta}{\mu}M\right)$$

Next… Coding

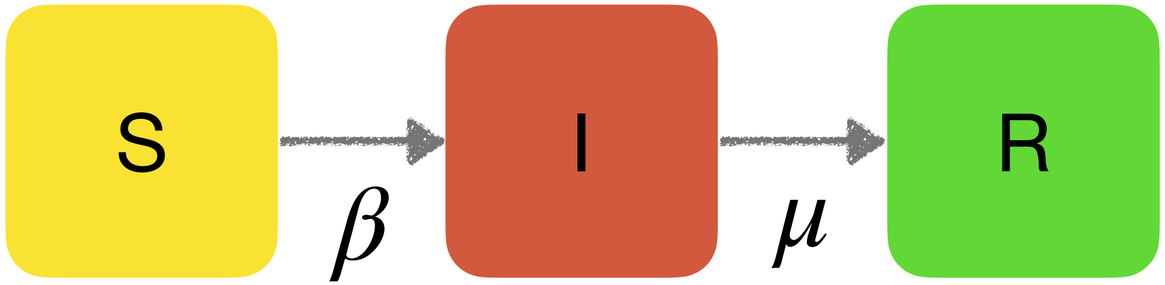
Deterministic vs stochastic models

- Deterministic models provide insights into the general behavior of the system (e.g. threshold effect)
- They represent a good way to study closed systems, and infer basic parameters
- They lack important details for realism:
  - **Stochasticity** (effects of chance)
  - The condition $R_0>1$ does not guarantee an outbreak to take off
  - **Discretization** (real epidemic variables are not continuous)

Stochastic numerical simulations

- Variables representing the compartments are discrete
- Transitions between compartments are simulated as random binomial or multinomial processes
- This is done through the use of **(pseudo) Random Number Generators**

Stochastic SIR model



- infection and recovery are **random processes**, like flipping coins
- probability of success: $\beta \Delta t I(t)/N$, $\mu \Delta t$
- number of trials: $S(t)$, $I(t)$
- number of successes: $\Delta S = Binom(S(t), \beta \Delta t I(t)/N)$,

$\Delta R = Binom(I(t), \mu \Delta t)$

Stochastic SIR model

- Fundamental **differences** with respect to the deterministic SIR:
  - ⋆ each realization of the model is different (many simulations are required, sometimes millions)
  - ⋆ the outcome of a simulation **depends on the initial conditions**, I_0
  - ⋆ the value of R_0 determines the probability of an outbreak to occur

$$p_{outbreak} = 1 - \frac{1}{R_0^{I_0}}$$